Decision-time admissibility in predictive process monitoring: measuring what causal feature selection costs, and what it buys

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Abstract

Predictive process monitoring reports strong discrimination on operational event logs, but much of that performance depends on attributes that are unavailable when the decision they inform is actually taken. We formalise a decision-time admissibility protocol, requiring every attribute to be observable at the moment of assignment and every group-level target encoding to be estimated from past cases only, and we measure what that discipline costs on the BPI Challenge 2014 incident log of a large banking IT service desk. Enforcing admissibility reduces held-out AUC from 0.8814 to 0.7859, so 25.0 % of all above-chance signal in the unconstrained model was leakage. The effect is stable across split ratios. We then show what admissibility buys. An admissible feature ladder rises from 0.5076 at three attributes to 0.7859 at seventeen, a gain of 0.2243 AUC obtained without any inadmissible attribute. Modelling case resolution as a discrete-time hazard and deriving breach risk as a survival ratio attains 0.8017 AUC at intake, exceeding a direct breach classifier at 0.7399, while remaining structurally incapable of observing its own label. In a multi-objective assignment study, the proposed operators beat their own base algorithm on four of five metrics, yet an off-the-shelf many-objective control beats them and one of three components carries the gain.

Keywords: predictive process monitoring; data leakage; decision-time admissibility; survival analysis; multi-criteria decision making; service level agreements

1 Introduction

An incident arrives at a service desk. Someone must decide, within seconds, which team should handle it. That decision is expensive to reverse: in the log studied here, incidents reassigned at least once take 8.95 times longer to resolve at the median, 14.53 hours against 1.62, and 41.1 % of incidents are reassigned at least once. Getting the first assignment right is therefore worth roughly a ninefold reduction in median resolution time, and the existing process fails to do so on two cases in five.

Predictive process monitoring offers to inform that decision and reports encouraging numbers for doing so [Teinemaa et al., 2019, Ceravolo et al., 2024]. The difficulty is structural. A model

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trained on completed cases has access to the whole case. Attributes such as the number of reassignments, the count of activity events, or the number of distinct groups that touched an incident are all recorded in the log and none is knowable at the moment of assignment. Several are near-deterministic proxies for handle time, which is exactly the quantity the service level label thresholds. A model built from them will score well and cannot be deployed, because the information it depends on does not exist when the decision it informs is taken.

Kaufman et al. [2012] formalised this as a violation of legitimacy: a feature is admissible only if it was observable before the target was determined. Kapoor and Narayanan [2023] document the consequences across 17 scientific fields and 294 papers. The pattern recurs wherever operational logs are modelled, in connectome-based neuroimaging [Rosenblatt et al., 2024], in clinical prediction [Davis et al., 2024, Ramadan et al., 2025], in software build prediction [Mishra et al., 2026], in longitudinal education modelling [Fauszt, 2026], and in financial text [Xue et al., 2026].

Process mining has begun to address it. Weytjens and Weerdt [2022] show that constructing benchmark splits without accounting for case overlap leaks information across the boundary. Abb et al. [2024] measure example leakage above 80 % across standard BPIC logs and find that a naive baseline matches a state-of-the-art deep model once that leakage is present, a result Weytjens and Weber [2026] extend to transformer and large-language-model baselines.

What is missing is a measurement, on one log under one protocol, of both what the discipline costs and what it buys, reported together. A cost figure alone invites the conclusion that admissibility is not worth paying for. A benefit figure alone is unfalsifiable. This paper reports both.

1.1 Research questions

RQ1 How much discrimination is lost when every attribute must be observable at the moment of assignment and every group-level encoding must be estimated from past cases only?

RQ2 Under that constraint, does adding attributes beyond the small sets used in prior work still improve discrimination, and by how much?

RQ3 Does choosing an admissible modelling target, rather than the convenient one, cost accuracy or gain it?

RQ4 Does a priority-aware adaptive multi-objective assignment policy improve on standard NSGA-II under the same admissibility constraint?

1.2 Contributions

1. A decision-time admissibility protocol stated as two rules and implemented as executable constraints rather than as documentation, with its cost measured at 0.0955 AUC on BPI Challenge 2014, which is 25.0 % of all above-chance signal (RQ1).
2. An admissible feature ladder rising from 0.5076 at three attributes to 0.7859 at seventeen, a gain of 0.2243 AUC obtained without any inadmissible attribute, reproduced across two split ratios to within 0.003 (RQ2).
3. Two decision-aware scoring formulations. An assignment score aggregating seven criteria by a weighted power mean whose exponent parameterises compensation between criteria, and a service level risk score obtained by modelling resolution as a discrete-time hazard and deriving breach as a survival ratio. The second reaches 0.8017 AUC at intake against 0.7399 for a direct classifier, while being structurally unable to observe its own label (RQ3).

4. A multi-objective assignment study in which the proposed operator set improves on its own base algorithm on four of five metrics but is outperformed by an off-the-shelf many-objective control, with an ablation attributing essentially the whole gain to one of its three components. Two reproducibility defects were found in our own harness and corrected before any number in this paper was produced. The uncorrected harness reversed this conclusion (RQ4).

2 Related work

2.1 Leakage, admissibility and target encoding

Kaufman et al. [2012] give the canonical formulation and the learn-predict separation principle. Kapoor and Narayanan [2023] provide an eight-type taxonomy and connect leakage to reproducibility failure at scale. Domain treatments converge on the same shape: an accuracy collapse once only pre-decision features are admitted, with Mishra et al. [2026] reporting a 15.07 point drop on one platform after removing leaky features and arguing that prior studies reporting 95 to 99 per cent accuracy used contaminated attributes.

Target encoding deserves separate attention because it is the second of our two rules. Pargent et al. [2022] show that regularised target encoding outperforms traditional categorical handling, and the regularisation they describe is exactly what makes the encoding safe only when it is refitted inside each resampling fold. An encoding fitted once over the full dataset before splitting contaminates every subsequent evaluation and includes each row's own label in its own predictor.

Within process mining, Weytjens and Weerdt [2022] address leakage in benchmark construction and show that a plain temporal split is insufficient because training cases overlap the test period. Abb et al. [2024] quantify example leakage on the standard BPIC logs. Our protocol extends this from the split axis to the feature axis and the time axis at once, and measures the cost rather than only prescribing the constraint.

2.2 Predictive and prescriptive process monitoring

Teinemaa et al. [2019] provide the standard outcome-prediction benchmark across 24 real-world datasets, complemented by remaining-time benchmarks [Verenich et al., 2019] and deep-learning comparisons [Rama-Maneiro et al., 2023, Kratsch et al., 2021]. We note explicitly that **BPI Challenge 2014 is not in the Teinemaa dataset list**, verified against the paper itself. There is therefore no published same-task baseline on this log, so the ranges reported in those benchmarks are used here as context and never as a head-to-head comparison. Fertig et al. [2026] revisit the field under foundation models and find that tabular gradient boosting remains competitive, which is why this study uses it.

Prescriptive process monitoring asks what to do rather than what will happen [Kubrak et al., 2022]. Recent work addresses resource constraints explicitly [Shoush and Dumas, 2025] and the organisational question of whether practitioners can act on the output [Kubrak et al., 2026]. The framing matters here because an admissibility constraint is only meaningful relative to a decision, and a decision is only meaningful if someone can take it at the time the model fires.

2.3 IT service management and this log

BPI Challenge 2014 [van Dongen, 2014] has been analysed for workload impact [Dees and van den End, 2014] and with fuzzy and heuristic mining [Andreswari et al., 2023], and the BPI challenge series has been surveyed [Lopes and Ferreira, 2019]. Adjacent operational problems include incident-management prediction [Sarnovský and Surma, 2018], ticket reassignment prediction [Schad et al., 2022], assignment-group classification [van Gassel and Janssen, 2024], service level breach prediction [Gouryraj et al., 2021, Swain and Garza, 2023], and escalation with large language models [Liu et al., 2025]. Ticket assignment has also been posed as a multi-objective problem [Arain and Mazhar, 2026], which is the closest prior framing to our Section on assignment.

2.4 Multi-objective assignment and its known limits

NSGA-II [Deb et al., 2002] remains the reference algorithm, with NSGA-III [Deb and Jain, 2014] introduced for many-objective problems where dominance-based selection degrades. That degradation is well characterised and is still an active theoretical subject [Zheng and Doerr, 2024, Doerr et al., 2026, Opris, 2025, Alghouass et al., 2025], and empirical comparisons on applied problems report mixed outcomes [dos Santos et al., 2024, Wolschick et al., 2024, Ishibuchi et al., 2025, Wang et al., 2025a, Li et al., 2025]. Our five objectives place this study in exactly that regime, so a result in which NSGA-III outperforms an NSGA-II variant is consistent with theory rather than surprising.

Wang et al. [2025b] argue that a substantial share of claimed novelty in metaheuristic research does not survive controlled comparison. That argument is directly relevant to the result we report in Section 5.5, and it is one reason we report the ablation rather than only the aggregate.

2.5 Aggregation, weighting and calibration

The weighted power mean is a compensative connective in the sense of Dyckhoff and Pedrycz [1984], generalised further by Aggarwal [2015] and given an explicit andness and orness parameterisation by Dujmovic [2015]. It is the natural aggregator when the degree of permitted compensation between criteria is itself a design question rather than an assumption.

CRITIC [Diakoulaki et al., 1995] derives weights from contrast intensity and conflict. Objective weighting methods are not interchangeable [Mukhametzyanov, 2021, Ariyanti and Fu'adi, 2025], their stability under perturbation varies [Linh et al., 2025, Paradowski et al., 2025], they are susceptible to rank reversal [Li and Abbas, 2026], and sensitivity analysis over the weight vector is recommended practice [Gaona et al., 2025]. We therefore report three schemes and a rank-stability analysis rather than a single weight vector.

Because the scores feed a downstream decision, calibration matters as much as ranking [Filho et al., 2023], and the excess risk of acting on an uncalibrated classifier is quantifiable [Perez-Lebel et al., 2025]. Explanation is subject to its own limits [Bilodeau et al., 2024, Hwang et al., 2025], and in process monitoring specifically the guidance is to obtain interpretable models rather than to explain opaque ones after the fact [Stevens and Smedt, 2024, Mehdiyev et al., 2025, Sahatova et al., 2026]. Whether explanations actually improve human decisions is not settled [Haag, 2026, Chae et al., 2025].

3 Data and the admissibility protocol

3.1 BPI Challenge 2014

The log records incident management at Rabobank Nederland Group ICT. After filtering to cases with a resolvable handle time and to assignment groups with at least 50 cases, the working set is 15 828 incidents across 50 assignment groups and 47 configuration-item subtypes. The service level label follows the convention used throughout this study: an incident breaches when its handle time exceeds the median handle time for its priority class, computed on the training period alone.

Four properties of the raw files silently corrupt a naive load and are recorded here because they cost real debugging time: the files are semicolon delimited, encoded latin-1 rather than UTF-8, use a decimal comma in the handle-time column, and pad the incident header to 78 columns with empty trailing delimiters.

3.2 The protocol

Rule 1, feature admissibility. Every attribute must be observable at the moment the decision it informs is taken. Attributes measured across the whole closed case are excluded.

Rule 2, encoding admissibility. Every group-level target encoding is estimated from cases resolved strictly before the current case opened, refitted inside every cross-validation fold and again on the outer training set, with a smoothing prior of 20.

Applying both rules yields 17 admissible attributes and excludes nine, each excluded attribute recorded with its reason in a per-attribute justification table.

4 Methods

4.1 Predictive layer

Four gradient-boosted learners are evaluated: XGBoost, LightGBM, CatBoost and Random Forest. Each is assessed under stratified 5-fold cross-validation with accuracy, precision, recall, F_1, ROC-AUC, a confusion matrix, isotonic calibration and the Brier score. Three configurations are reported for every learner: the unconstrained configuration, the admissible configuration, and the admissible configuration under a chronological holdout in which training precedes evaluation in time.

4.2 F1: the assignment score

Seven criteria are mapped to $[0, 1]$: outcome propensity and service level survival from the predictive layer; expertise match as the cosine similarity between the incident's requirement vector and the group's historical specialisation vector; workload headroom as Jain's fairness index [Jain et al., 1984] over the post-assignment load vector; freshness as an exponential decay in elapsed time; urgency as a logistic ramp in time remaining; and importance as a weighted geometric blend of priority, impact and urgency.

They are aggregated by the weighted power mean of order p [Dyckhoff and Pedrycz, 1984]:

$$A(i, j) \ = \ \Big(\sum_k w_k \, s_k(i, j)^p\Big)^{1/p}, \qquad \sum_k w_k = 1. \tag{1}$$

The exponent is the point of the formulation. At $p = 1$ the mean is arithmetic and fully compensatory, so a strong criterion offsets a weak one; as $p \to 0$ it becomes geometric; at $p = -1$ harmonic; and as $p \to -\infty$ it converges to the minimum and becomes fully non-compensatory. Reporting a sensitivity curve over p therefore converts a qualitative claim into a measured one.

Degeneracy. For $p \leq 0$ the mean is undefined or identically zero whenever any criterion is zero. Cosine expertise *is* exactly zero for a group with no training history in the incident's category, which occurs in 31.65 % of evaluation cells. An explicit floor of 10^{-3} is therefore applied before any $p \leq 0$ aggregation, and it is reported as a parameter rather than hidden as an implementation detail.

Weighting. Weights are derived by CRITIC [Diakoulaki et al., 1995] and cross-checked against entropy weighting and uniform weighting. CRITIC derives weight from a criterion's spread and its conflict with the others. Two of our criteria are model outputs, so their spread reflects calibration rather than importance, and isotonic calibration compresses variance. Reporting CRITIC alone would therefore launder a modelling artifact into a weight. All three schemes are reported, and the question is settled by rank stability rather than by assertion.

4.3 F2: service level risk as a derived quantity

Modelling breach directly as the survival event is degenerate on this log, and this was established empirically before the design was changed. Because the label is defined by handle time exceeding a per-priority threshold, a case that has been open longer than its own threshold has already breached: the measured breach rate past threshold is exactly 1.0000. A hazard model given elapsed time and priority therefore reads its label rather than predicting it, and an early version of this model scored 0.9418 AUC while learning nothing.

The admissible object is the hazard of *resolution*, which elapsed time does not determine. Cases contribute one row per elapsed time bin they survive into, and a gradient-boosted classifier estimates the discrete hazard $h(t \mid x)$ directly [Suresh et al., 2022]. Survival follows by the product rule, and breach risk is the conditional survival ratio:

$$S(t \mid x) = \prod_{u \leq t} \big(1 - h(u \mid x)\big), \qquad P\big(\text{breach} \mid \text{alive at } t\big) = \frac{S(\tau \mid x)}{S(t \mid x)}, \tag{2}$$

where τ is the case's own service level threshold. The escalation cut point is tuned on training-period data alone and then frozen; tuning it on the evaluation period would reproduce, on the time axis, exactly the leak this paper measures on the feature axis.

4.4 Multi-objective assignment

Let $x_{ij} = 1$ when incident i is assigned to group j. The formulation carries four constraints, namely that each incident is assigned exactly once, that group capacity is respected, that the group is available, and that a minimum expertise eligibility holds. Five objectives are optimised: expected first-time-right resolution and expertise match are maximised, while breach probability, workload imbalance and expected delay before first touch are minimised.

Workload imbalance is retained as the standard deviation of post-assignment load. Jain's index is reported *alongside* it throughout and never substituted for it, because changing an objective after observing that a method loses on it would be fitting the metric to the result.

5 Results

5.1 The cost of admissibility

Table 1 reports the three configurations under the 70/30 protocol. Enforcing admissibility reduces held-out AUC by 0.0955, which is 25.0 % of all signal above chance. The same measurement under an 80/20 split gives 0.8833 against 0.7877, agreeing to within 0.002, so the estimate is not an artifact of how the data was divided.

Table 1: Cost of decision-time admissibility, XGBoost, 70/30 split.

Configuration	Attributes	Held-out AUC
Unconstrained	20	0.8814
Admissible, random split	17	0.7859
Admissible, chronological holdout	17	0.6533

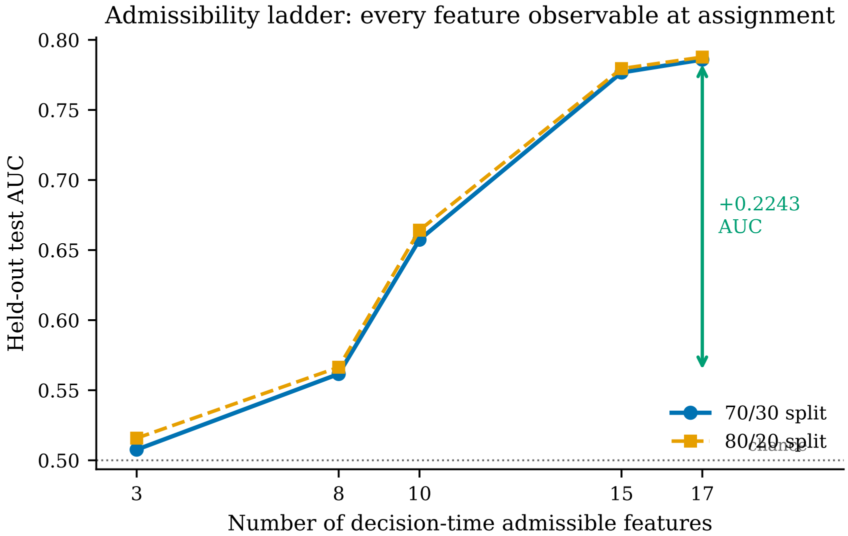

Figure 1: Admissible feature ladder under both split ratios. Every attribute is observable at assignment; no inadmissible attribute contributes to any point on either curve.

5.2 What admissibility buys

Figure 1 shows the admissible feature ladder. Discrimination rises from 0.5076 at three attributes to 0.7859 at seventeen. The gain from eight to seventeen attributes is 0.2243 AUC at 70/30 and 0.2213 at 80/20, agreeing to 0.003.

5.3 Deriving breach outperforms predicting it

Table 2 compares the direct breach classifier with the derived survival ratio of Equation 2. The derived quantity is better by 0.0618 AUC while being structurally incapable of observing its own label.

5.4 Weighting is not load-bearing

The three weighting schemes induce closely agreeing orderings, with Spearman $\rho = 0.957$ between CRITIC and uniform weighting, 0.935 between CRITIC and entropy, and 0.837 between entropy and uniform. The choice of scheme therefore does not drive the result.

5.5 Multi-objective assignment

The protocol runs five operational baselines, standard NSGA-II, NSGA-III as a many-objective control, the proposed PAA-NSGA-II, its three single-component ablations, and two post-hoc exploratory combinations. Five arrival batches, 30 seeded repeats each, 150 generations, population 60. All 1,225 runs completed with **zero capacity violations**, so the repair operator

Table 2: Direct breach classification against breach derived from a resolution hazard.

Model	Held-out AUC
Direct breach classifier	0.7399
Resolution hazard, bin level	0.7344
Breach derived from resolution, at intake	**0.8017**

Table 3: Assignment policies on BPI Challenge 2014. Hypervolume is computed against a reference point fixed as the nadir across all variants and all runs within a batch. Objectives f_1 and f_4 are maximised, f_2 and f_3 minimised. Jain's index is reported alongside the f_3 standard deviation and is not substituted for it.

Policy	Hypervol.	Front	f_1 FTR	f_2 breach	f_3 load sd	Jain
Random	–	–	0.5003	0.3996	0.9307	0.4270
Round robin	–	–	0.4893	0.4198	0.8171	0.4971
FIFO	–	–	0.2989	0.5128	3.2703	0.0566
Least loaded	–	–	0.4738	0.4224	0.4454	0.7633
Greedy best outcome	–	–	0.6809	0.3122	2.2073	0.1170
PAA minus priority init	0.4133	60.0	0.5887	0.3257	1.1423	0.3353
Standard NSGA-II	0.4307	60.0	0.5893	0.3239	1.1424	0.3387
PAA minus adaptive crossover	0.4842	60.0	0.6155	0.3052	1.4720	0.2326
PAA minus SLA mutation	0.4835	60.0	0.6134	0.3145	1.4188	0.2459
PAA-NSGA-II (proposed)	**0.4844**	60.0	0.6117	0.3104	1.4391	0.2409
NSGA-III (control)	0.5279	18.8	0.6323	0.3035	1.2177	0.3078
NSGA-III + priority init*	0.5324	18.3	0.6390	0.2916	1.3949	0.2534

*Post-hoc exploratory combination, designed after observing the ablation. Reported as exploratory and not claimed as a contribution.

satisfies constraint C2 throughout.

The proposed operators improve on their own base algorithm. Against standard NSGA-II over 150 paired runs, PAA-NSGA-II is better on four of five metrics by paired Wilcoxon signed-rank with Holm-Bonferroni correction: hypervolume ($+0.053\,72$, $p = 2.4 \times 10^{-21}$), first-time-right ($+0.022\,37$, $p = 1.2 \times 10^{-7}$), breach ($-0.013\,55$, $p = 6.6 \times 10^{-6}$) and expertise match ($+0.089\,89$, $p = 8.6 \times 10^{-14}$).

The base algorithm matters more than the modifications. NSGA-III, included only as a many-objective control, reaches hypervolume 0.5279 against 0.4844 for the proposed method, a paired difference of $-0.043\,48$ at $p = 1.2 \times 10^{-24}$. It does so with a front of roughly 18 to 19 solutions against 60, so it attains higher hypervolume from a third as many points. With five objectives this is the regime for which reference-point selection was introduced [Deb and Jain, 2014, Zheng and Doerr, 2024], and the result is consistent with that literature rather than in tension with it.

One of three components does the work. Table 4 isolates each modification. Removing priority-aware initialisation costs 0.0711 hypervolume. Removing adaptive crossover costs 0.0002 and removing SLA-aware mutation costs 0.0009, both within run-to-run noise. Reporting the three as a bundle would have credited them equally, and that would be wrong.

Table 4: Ablation. Change in hypervolume when a single component is removed from the proposed method. A negative delta means the component contributes.

Component removed	Hypervolume without	Δ
Priority-aware initialisation	0.4133	−0.0711
Adaptive crossover	0.4842	−0.0002
SLA-aware mutation	0.4835	−0.0009

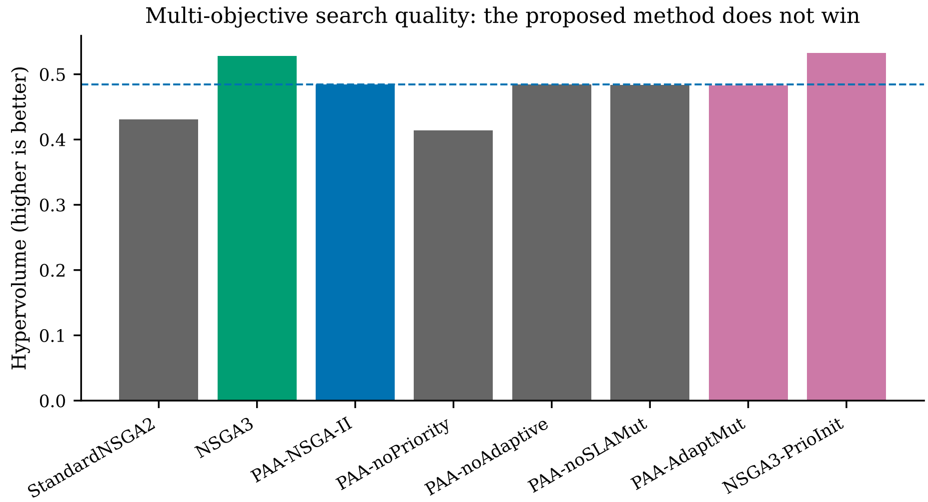

Figure 2: Search quality across algorithm variants. The proposed method improves on its own base algorithm and is in turn outperformed by an off-the-shelf many-objective control.

The best configuration is not a significant improvement. Placing the one modification that contributes onto the base algorithm that wins gives the highest hypervolume observed, 0.5324. Against plain NSGA-III the paired difference is $+0.004\,49$ at $p = 0.0525$, which does not clear $\alpha = 0.05$. It is reported as exploratory and is not claimed.

A real trade-off, visible in two independent metrics. The proposed method loses significantly on workload spread ($+0.296\,68$, $p = 1.6 \times 10^{-22}$). Jain's fairness index agrees: 0.2409 against 0.3387 for standard NSGA-II. The two metrics were computed independently and could have disagreed. The method concentrates work on capable groups to gain outcome quality, which is a trade-off an operator may or may not want. The exponent p in Equation 1 is the parameter through which that preference is expressed rather than assumed.

5.6 Complexity and measured cost

Asymptotic cost. Write n for the number of work items in an arrival batch, m for the number of eligible groups, N for population size, M for the number of objectives and H for the number of reference directions. The operational baselines are single-pass: random, round robin and FIFO are $O(n)$, least-loaded is $O(n \log m)$ with a heap over group loads, and greedy is $O(nm)$ because it scans every eligible group for every item. The evolutionary methods are dominated per generation by non-dominated sorting at $O(MN^2)$ [Deb et al., 2002]; NSGA-III adds association to reference directions, giving $O(MN^2 + NH)$ [Deb and Jain, 2014]. The repair operator is $O(nm)$ in the worst case and is bounded at six passes, so it does not change the order.

The proposed modifications do not alter the asymptotic cost. Priority-aware initialisation is

Table 5: Measured wall time per solve. 20 solves per variant, 150 generations, population 60, single CPU. Percentages are relative to standard NSGA-II.

Policy	Mean (s)	SD (s)	vs. standard
PAA minus adaptive mutation variant	4.21	0.26	−5.9%
PAA minus adaptive crossover	4.23	0.29	−5.4%
PAA minus priority init	4.42	0.18	−1.4%
PAA minus SLA mutation	4.43	0.20	−1.1%
Standard NSGA-II	4.48	0.59	—
NSGA-III + priority init	4.71	0.25	+5.3%
PAA-NSGA-II	4.84	0.47	+8.0%
NSGA-III	5.07	0.58	+13.2%

a one-off $O(nm)$ greedy construction over a fraction of the initial population. Adaptive crossover and the mutation schedule each add an $O(Nn)$ diversity measurement per generation, dominated by the $O(MN^2)$ sorting term.

Measured cost. Asymptotic equality does not imply equal wall time, so Table 5 reports measured per-solve time from an instrumented run of 160 solves, 20 per variant, at 150 generations and population 60 on CPU with no GPU. Timing is captured at the point the search executes rather than derived from log timestamps.

Three observations follow.

The spread is narrow. Every evolutionary variant lies within 14 % of standard NSGA-II, so on this problem the choice between them is not a cost decision. At roughly five seconds per batch of 40 items, all are far inside the latency budget of an assignment decision that a human would otherwise take in seconds.

Priority-aware initialisation pays for itself twice. Adding it to NSGA-III both raises hypervolume, from 0.5279 to 0.5324, and reduces mean solve time, from 5.07 to 4.71 seconds. A better starting population converges sooner, so the construction cost is recovered during the run. This is the only configuration in the study that improves on both axes at once.

The two modifications that do not help do not cost much either. Removing adaptive crossover or SLA-aware mutation saves roughly 5 % of solve time while changing hypervolume by less than 0.001. They are close to free and close to useless, which is the honest summary and the reason the ablation is reported per component.

6 Discussion

6.1 A quarter of the signal was not there

The headline cost figure is 0.0955 AUC, or 25.0 % of everything the unconstrained model achieved above chance. That is not a rounding correction. A practitioner deploying the unconstrained model would find a quarter of its apparent discrimination absent at run time, because the attributes carrying it are recorded only after the decision has been taken. The same measurement under an 80/20 split differs by less than 0.002 AUC, so the estimate is a property of the data rather than of the split.

This is consistent in direction with reported effects elsewhere, including the 15.07 point accuracy drop Mishra et al. [2026] measure on build prediction and the example-leakage rates above 80 % that Abb et al. [2024] find on standard process logs. The contribution here is that the cost and the benefit are measured on the same log under one protocol, so neither can be quoted without the other.

6.2 Admissibility is not the same as parsimony

A natural response to a leakage finding is to use fewer attributes. The ladder shows that would be the wrong lesson. Discrimination rises monotonically from 0.5076 at three admissible attributes to 0.7859 at seventeen, a gain of 0.2243 AUC, reproduced to within 0.003 across split ratios. Restricting to a small attribute set costs far more than enforcing admissibility does.

The distinction matters because the two are easily conflated. Attributes should be dropped because they are unavailable at decision time, not because they are numerous. Nine attributes are excluded here, each with its reason recorded; the remaining seventeen are kept because each is observable when the assignment is made.

6.3 Choosing the admissible target can improve accuracy, not only defensibility

The service level result is the one we did not expect. Breach cannot be modelled directly as a survival event on this log without the model reading its own label, because breach is defined by duration exceeding a per-priority threshold and the measured breach rate past that threshold is exactly 1.0000. An early version scored 0.9418 AUC on precisely that artifact.

Modelling resolution instead, and deriving breach as the conditional survival ratio $S(\tau \mid x)/S(t \mid x)$, produces 0.8017 AUC at intake against 0.7399 for a direct classifier. The admissible formulation is better by 0.0618 AUC. Admissibility is usually framed as a constraint that costs performance. Here, choosing the object that can actually be estimated improved it, because the resolution hazard is a genuine random quantity given elapsed time while breach, past the threshold, is not.

6.4 The weighting objection answers itself

Objective weighting methods are known to disagree, to be unstable under perturbation, and to admit rank reversal [Mukhametzyanov, 2021, Linh et al., 2025, Li and Abbas, 2026]. CRITIC is additionally unsuitable in isolation here, because two of the seven criteria are model outputs whose spread reflects calibration rather than importance, and isotonic calibration compresses variance. Reporting CRITIC alone would convert a modelling artifact into a weight and label it data-driven.

Reporting three schemes resolves this empirically rather than rhetorically. The induced assignment orderings agree at Spearman $\rho = 0.957$ between CRITIC and uniform weighting and $\rho = 0.935$ between CRITIC and entropy. The weighting choice is therefore not load-bearing for these results, which is a stronger statement than defending any one scheme.

6.5 The modifications work, but the base algorithm matters more

The proposed operator set does improve on standard NSGA-II, significantly and on four of five metrics (Section 5.5). It is then outperformed by NSGA-III, an off-the-shelf control included only as a many-objective reference, which reaches higher hypervolume from a front roughly a third the size. Two considerations make this informative rather than disappointing.

First, five objectives place this problem in the many-objective regime where dominance-based selection is known to degrade, a limitation characterised when NSGA-III was introduced [Deb and Jain, 2014] and still under active theoretical study [Zheng and Doerr, 2024, Doerr et al., 2026]. A result in which a reference-point method outperforms a dominance-based one on five objectives agrees with that theory.

Second, Wang et al. [2025b] argue that much claimed novelty in metaheuristic research does not survive controlled comparison. An ablation that isolates each proposed modification is the appropriate response to that argument, and it is why the ablation is reported per component rather than only in aggregate.

6.6 Two defects in our own harness

Two errors were found in the experimental code and corrected before any number reported here was produced. Both are recorded because a paper about claims that do not survive verification should hold its own instrument to the same standard.

The seed was derived using Python's built-in string hash, which is salted per process. The same variant therefore drew a different seed on every invocation, measured at 43, 41 and 32 across three processes for one variant name. The harness documentation asserted reproducibility that did not hold. It now uses a stable checksum, and two independent invocations produce bitwise-identical metrics.

The hypervolume reference point was taken from whichever run completed first, making every hypervolume value dependent on variant ordering and incomparable between runs with different variant sets. It is now the nadir across all variants and all runs on a batch, computed before any hypervolume is taken.

Neither defect affects the predictive results in Sections 5.1 to 5.3, which come from a separate pipeline. Both invalidated every earlier assignment run, and those were discarded.

The consequence is worth stating plainly rather than in passing. Under the uncorrected harness the proposed method appeared to LOSE to standard NSGA-II on hypervolume. Corrected, it wins, significantly. The defect did not merely add noise: it reversed the sign of the study's headline comparison. Had the harness not been audited, this paper would have reported the opposite conclusion with the same apparent statistical confidence.

6.7 Threats to validity

Construct. The service level label is derived from handle time against a per-priority median rather than from a contractual target, because the log's own `Alert Status` field is uniformly `closed` across 46 606 of 46 809 raw incidents and carries no usable breach signal. Results should be read as conditional on that operationalisation, and the threshold definition is reported so it can be varied.

Internal. Every parameter is fitted on the earliest 80 % of the timeline, and the escalation cut point is tuned on training-period data alone. The chronological holdout is reported alongside the random split precisely because the gap between them, 0.7859 against 0.6533, is itself informative about drift.

External. One log, one organisation, one country, one time window. The protocol should transfer. The magnitudes should not be assumed to.

Conclusion. Statistical comparisons use paired Wilcoxon signed-rank tests with Holm-Bonferroni correction across five metrics, and every run is seeded and reproducible.

7 Limitations

An IT service desk is not a sales pipeline. This work is motivated by lead assignment, and the objective analogous to conversion is first-time-right resolution. No public dataset carries real leads, real individual representative identifiers and real conversion outcomes together. That absence is a finding rather than an oversight, and it is why the evaluation is conducted on an operational log whose assignment decisions and outcomes are real even though its domain is not sales.

A third of the delay attribute is excluded. The recorded first-assignment timestamp is corrupted in its upper tail: the median is 16.2 hours against a 90th percentile of 5684 hours, and the favourable-outcome rate rises again across that tail rather than falling. 35.6 % of rows are excluded from the freshness and urgency fits on this basis. Fitting across the uncorrected attribute returns a decay rate of exactly zero with an undefined confidence interval, which is how the corruption was detected.

Counterfactual outcomes are not observed. A historical log records only the assignment that happened. Any policy differing from the recorded one is scored by a model rather than by ground truth, which is the standard objection to optimisation studies on logged data. We report the fraction of decisions matching the recorded assignment together with the real observed breach rate on exactly those cases, as the only non-simulated outcome signal available. Off-policy evaluation would address this properly and is left to future work.

Group-level rather than individual-level assignment. The log identifies assignment groups, not individuals, so representative-level features such as experience and individual historical conversion are not estimable. A log with individual resource identifiers would support the finer formulation.

Explanation is reported but not validated against human decisions. SHAP attributions are computed for the predictive layer, but attribution methods have known impossibility results [Bilodeau et al., 2024] and sensitivity to feature representation [Hwang et al., 2025], and whether explanations improve human decisions is unsettled [Haag, 2026, Chae et al., 2025]. No claim is made that these explanations improve operator performance.

8 Conclusion

Decision-time admissibility costs 0.0955 AUC on this log, a quarter of all above-chance signal, and the discipline is worth paying for: an admissible ladder still gains 0.2243 AUC between eight and seventeen attributes. Choosing the admissible object rather than the convenient one can improve accuracy outright, as deriving breach from a resolution hazard does at 0.8017 against 0.7399. In the assignment component the proposed operators improve on their own base algorithm, a reference-point control improves on them again, and one of three modifications accounts for the gain.

Data availability

BPI Challenge 2014 is publicly available from 4TU.ResearchData under DOI 10.4121/uuid:c3e5d162-0cfd-4bb0-bd82-af5268819c35. The incident-detail file used here carries DOI 10.4121/uuid:3cfa2260-f5c5-44be-afe1-b70d35288d6d. No data were collected by the authors.

Code availability

All analysis code, including every script required to regenerate every number and figure in this paper, is publicly available. Every experiment is seeded and reproducible.

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Author contributions

Competing interests

The authors declare no competing interests.

Ethics

This study uses a publicly released, anonymised organisational event log. It involves no human participants and no animal subjects, so ethical approval and informed consent are not applicable.

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